A PINCER PICTURE FOR THE MUTUAL CORRELATED AGREEMENT THRESHOLD

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Abstract. For Reed–Solomon codes with $k \le 2^{40}$ and $|\mathbb{F}| < 2^{256}$ — the parameter box singled out by the Proximity Prize — we give a sharpened pincer picture for the grand mutual correlated agreement threshold $\delta^*_C(2^{-128})$. For $|\mathbb{F}| < 2^{128}$ the threshold is exactly 0. For $|\mathbb{F}| \ge 2^{128}$ and the four challenge rates, ordinary locator-prefix floors already force plain correlated-agreement failure from $\delta = 1-\rho-s_\rho/n$ up to capacity, where $s_\rho = \lceil \alpha_\rho n \rceil$ and $\alpha_{1/2} = 1/300$, $\alpha_{1/4} = 3/1000$, $\alpha_{1/8} = 41/20000$, $\alpha_{1/16} = 3/2500$; in particular the old near-capacity problem has a negative answer on an explicit upper slice of its band. On the safe side, a self-contained theorem gives MCA with error $(\lfloor \delta n \rfloor + 1)/q$ below one third of the distance, and the unique-decoding proximity gap of Ben-Sasson–Carmon–Ishai–Kopparty–Saraf raises the safe frontier to one half of the distance whenever $|\mathbb{F}| \ge 2^{128} n$. Above half the distance, the extra mutual-only layer is reduced to explicit finite objects: tangent slopes, residual Reed–Solomon shortenings of dimension at most $\max(0, 2r-(n-k))$, and equivalently a doubled-radius pair-list obstruction. The half-distance wall is sharp, and the remaining threshold problem is reduced to a middle plain-CA band together with those residual shortening or pair-list images.

1. Introduction and main result

The Proximity Prize [PP26] and the challenge collection of Arnon, Boneh and Fenzi [ABF26] ask for the largest proximity parameter δ at which mutual correlated agreement (MCA) holds for a Reed–Solomon code with error at most a stated target, and single out one parameter box: *"we are mostly interested in determining δ^*_C when $L \subseteq \mathbb{F}$ is a smooth domain, $k \le 2^{40}$, and $|\mathbb{F}| < 2^{256}$."* This paper gives a pincer answer for that box: a certified safe region, a certified unsafe region strictly below capacity, and an exact formulation of the residual kernel. Write $C = \mathrm{RS}[\mathbb{F}, D, k]$ for the code of polynomials of degree less than k evaluated on a domain D of size n, $\rho = k/n$, $q = |\mathbb{F}|$, and let $\delta^*_C(\varepsilon^*)$ be the threshold of Definition 2.1 below, at the grand challenge target $\varepsilon^* = 2^{-128}$.

Theorem 1.1. *Let $q = |\mathbb{F}| < 2^{256}$, let $D \subseteq \mathbb{F}^\times$ be a coset of a cyclic subgroup of even order n, and let $C = \mathrm{RS}[\mathbb{F}, D, k]$ with $\rho = k/n \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$, $k \le 2^{40}$ and $n \ge 2^{15}$. Put*

$$\alpha_\rho := \begin{cases} 1/300, & \rho = \frac{1}{2}, \\ 3/1000, & \rho = \frac{1}{4}, \\ 41/20000, & \rho = \frac{1}{8}, \\ 3/2500, & \rho = \frac{1}{16}, \end{cases} \qquad s_\rho := \lceil \alpha_\rho n \rceil,$$

and put $g_\rho := 2^{-9}$ for $\rho \ge \frac{1}{8}$ and $g_\rho := 2^{-10}$ for $\rho = \frac{1}{16}$.

(1) *If $q < 2^{128}$, then $\delta^*_C(2^{-128}) = 0$.*
(2) *If $q \ge 2^{128}$, then*
$$\delta^*_C(2^{-128}) \le 1 - \rho - \frac{s_\rho}{n}.$$
More strongly, $\varepsilon_{\mathrm{ca}}(C, \delta) > 2^{-128}$, and hence $\varepsilon_{\mathrm{mca}}(C, \delta) > 2^{-128}$, for every $\delta \in [1 - \rho - s_\rho/n, 1 - \rho)$. Since $s_\rho/n \ge \alpha_\rho > g_\rho$, this gives a negative answer to the original plain-CA question on the nonempty upper slice $[1 - \rho - s_\rho/n,\ 1 - \rho - g_\rho)$ of its band.
(3) *If moreover $q \ge 2^{128} n$, then unconditionally $\frac{1}{n}\left\lfloor \frac{n-k}{3} \right\rfloor \le \delta^*_C(2^{-128}) \le 1 - \rho - s_\rho/n$, a determination within a multiplicative factor smaller than three; and, using additionally*

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the unique-decoding proximity gap of [BCIKS20], $\frac{1}{n}\lfloor\frac{n-k}{2}\rfloor \le \delta^*_C(2^{-128}) \le 1-\rho-s_\rho/n$, *a determination within a factor smaller than two.*

(4) *If* $2^{128} \le q < 2^{128}n$, *the same upper bound holds, and the unconditional lower bound is* r^*/n *with* $r^* := \min\big(\lfloor (n-k)/3\rfloor, \lfloor 2^{-128}q\rfloor - 1\big)$, *which degrades to* 0 *as* q *approaches* 2^{128}*; no factor claim is made in this thin regime.*

Every ingredient except the single import in (3) is proved here from first principles: a graded locator-prefix pigeonhole (Lemma 3.1), a deep-point list-to-correlated-agreement conversion (Theorem 3.2), an ordinary locator-prefix specialization that resolves the top of the plain-CA band (Corollary 3.4), a deep-regime theorem showing MCA *holds* with error $(\lfloor \delta n\rfloor + 1)/q$ below one third of the minimum distance for every linear code (Theorem 4.1), and a theorem showing that below one half of the distance mutual and plain correlated agreement coincide up to an explicit tangent term (Theorem 4.2). No value-set counting, exponential-sum estimate, or equidistribution input is used. The one import, [BCIKS20, Thm. 4.1], is a published theorem, so clause (3) stands on the literature as stated; we track it separately only to delimit the self-contained core. Section 7 then proves two barriers — an entropy envelope past which no prefix floor can certify failure (Proposition 7.1), and a ceiling of $\frac{1}{k}$ on the error certifiable through the conversion (Proposition 7.2) — and states the residual kernel after the two broad band questions are sharpened: a smaller plain-CA interval (Problem 7.4) and the residual shortening or pair-list obstruction (Problem 7.5). Remarks 3.7–3.8 complete the parameter box: characteristic-two additive smooth domains and odd 3-smooth orders satisfy the same baseline bounds, small n down to 2^{12} retains a nonvacuous cap of gap $3/n$, and rates outside the four challenge values are covered by the same finite criterion.

The upper bounds sharpen, and the composition follows, a line of work on capacity-edge obstructions: the conditional universal cap of [Cho26b] (gap 2^{-11} under $|\mathbb{F}| \ge 2n$), the locator framework of [Cho26a], and the conversion perspective of [CS25]; the strongest known positive results for plain correlated agreement are [BCIKS20, BCHKS25], reaching the Johnson radius. A companion manuscript [Cho26b] develops the deployed-parameter refinements (subfield-pigeonhole floors widening the unsafe frontier to gap $\approx 2^{-4.96}$ at $n = 2^{21}$, circle and genus-one rows, explicit witnesses) and the structured form of the remaining problem; the present paper is the minimal self-contained core plus the residual reductions needed to state the current picture.

2. Definitions

Throughout, $\mathbb{F}$ is a finite field, $q = |\mathbb{F}|$, $D \subseteq \mathbb{F}$ has $|D| = n$, and $C = \mathrm{RS}[\mathbb{F}, D, k] := \{p|_D : p \in \mathbb{F}[X],\ \deg p < k\}$ with $1 \le k < n$, so C is linear of dimension k and minimum Hamming weight $n-k+1$. For $u, v \in \mathbb{F}^D$, $\mathrm{dist}(u, v)$ is the relative Hamming distance, and for a set $C' \subseteq \mathbb{F}^D$, $\mathrm{dist}(u, C') := \min_{c\in C'} \mathrm{dist}(u, c)$ and $\Lambda(C', \delta, u) := \{c \in C' : \mathrm{dist}(u, c) \le \delta\}$. For a pair $(f_1, f_2) \in \mathbb{F}^D \times \mathbb{F}^D$ the column distance to $C^{\equiv 2} := C \times C$ is $\mathrm{dist}_2\big((f_1, f_2), C^{\equiv 2}\big) := \min_{c_1,c_2\in C} \frac{1}{n}\#\{j : (f_1(j), f_2(j)) \ne (c_1(j), c_2(j))\}$. All slopes are finite: lines are $\gamma \mapsto f_1 + \gamma f_2$, $\gamma \in \mathbb{F}$.

Fix $\delta \in (0, 1)$. A slope γ is *δ-CA-bad* for the pair (f_1, f_2) if the point $f_1 + \gamma f_2$ is δ-close to C while the pair is δ-far in columns, $\mathrm{dist}_2\big((f_1, f_2), C^{\equiv 2}\big) > \delta$. Following [ABF26, Defs. 4.1, 4.3], a slope γ is *δ-MCA-bad* for (f_1, f_2) if there exist $c \in C$ and $S \subseteq D$ with $|S| \ge (1-\delta)n$ and $(f_1 + \gamma f_2)|_S = c|_S$, such that *no* pair $(c_1, c_2) \in C^{\equiv 2}$ has $f_1|_S = c_1|_S$ and $f_2|_S = c_2|_S$; that is, some witness of the point's proximity fails to extend mutually to the pair. The two errors are

$$\varepsilon_{\mathrm{ca}}(C, \delta) := \max_{(f_1,f_2)} \frac{1}{q}\#\{\gamma\ \delta\text{-CA-bad}\}, \qquad \varepsilon_{\mathrm{mca}}(C, \delta) := \max_{(f_1,f_2)} \frac{1}{q}\#\{\gamma\ \delta\text{-MCA-bad}\}.$$

Definition 2.1. For a target $\varepsilon^* \in (0, 1)$, the grand MCA threshold is $\delta^*_C(\varepsilon^*) := \sup\{\delta \in (0, 1-\rho) : \varepsilon_{\mathrm{mca}}(C, \delta) \le \varepsilon^*\}$, with $\sup \emptyset := 0$.

The restriction to sub-capacity radii matches the challenge convention of [ABF26, PP26]; at $\delta \ge 1-\rho$ every word is δ-close to C by interpolation and the threshold question changes character. We record the comparison of the two errors.

Lemma 2.2. *Every δ-CA-bad slope is δ-MCA-bad; hence $\varepsilon_{\mathrm{ca}}(C,\delta) \le \varepsilon_{\mathrm{mca}}(C,\delta)$.*

Proof. If γ is CA-bad, then $\mathrm{dist}(f_1+\gamma f_2, C) \le \delta$, so a witness (c,S) with $|S| \ge (1-\delta)n$ exists; and $\mathrm{dist}_2((f_1,f_2), C^{\equiv 2}) > \delta$ means no set of $\ge (1-\delta)n$ coordinates carries a pair explanation, in particular S does not. So γ is MCA-bad. □

We use the standard entropy estimates for binomial coefficients, with $\mathrm{H}_2(x) = -x\log_2 x - (1-x)\log_2(1-x)$: for integers $0 < m < N$,

$$2^{N\mathrm{H}_2(m/N)}/(N+1) \;\le\; \binom{N}{m} \;\le\; 2^{N\mathrm{H}_2(m/N)}. \tag{1}$$

Indeed, with $p = m/N$, expanding $1 = \sum_i \binom{N}{i} p^i(1-p)^{N-i}$ gives the upper bound by keeping the term $i = m$, which equals $\binom{N}{m}2^{-N\mathrm{H}_2(p)}$; and since the term ratios $\binom{N}{i+1}p^{i+1}(1-p)^{N-i-1}/\binom{N}{i}p^i(1-p)^{N-i} = (N-i)p/((i+1)(1-p))$ exceed 1 for $i < m$ and are below 1 for $i \ge m$, the $i = m$ term is the largest of the $N+1$ terms, giving the lower bound.

3. The unsafe side

The failure mechanism has two independent parts: a pigeonhole construction of a single received word carrying an enormous list at an agreement slightly above k, and a conversion of any such list into correlated-agreement failure. Neither part uses any arithmetic property of $\mathbb{F}$ beyond its size.

Lemma 3.1 (graded prefix floor)**.** *Let $\varphi \in \mathbb{F}[X]$ have degree $c \ge 1$ and suppose the fibers of φ inside D are complete and of uniform size c: every nonempty set $\varphi^{-1}(y) \cap D$, $y \in Q := \varphi(D)$, has exactly c elements. Put $N := n/c$, let $B \subseteq \mathbb{F}$ be any subfield with $D \subseteq B$ and $\varphi \in B[X]$ (always $B = \mathbb{F}$ works), and let $1 \le K \le n$ and $\lceil K/c\rceil \le m \le N$, $w := m - \lceil K/c\rceil$. Then some B-valued word $U\colon D \to B$ satisfies*

$$\left|\Lambda\big(\mathrm{RS}[\mathbb{F},D,K],\, 1-\tfrac{mc}{n},\, U\big)\right| \;\ge\; \binom{N}{m}\Big/|B|^{w}.$$

Proof. For each m-subset $M \subseteq Q$ put $\Lambda_M := \prod_{b\in M}(\varphi - b) = \sum_{j=0}^{m}(-1)^j e_j(M)\varphi^{m-j}$, where e_j is the j-th elementary symmetric function; $\Lambda_M \in B[X]$ has degree cm and vanishes at $x \in D$ exactly when $\varphi(x) \in M$, i.e. at exactly cm points by fiber completeness. For $z \in B^w$ let $U_z := \big(\varphi^m + \sum_{j=1}^{w} z_j\varphi^{m-j}\big)\big|_D$, a B-valued word. By pigeonhole over the prefix map $M \mapsto z(M) := \big((-1)^j e_j(M)\big)_{j\le w} \in B^w$, some z^* is attained by at least $\binom{N}{m}/|B|^w$ sets M; for each such M set $c_M := U_{z^*} - \Lambda_M|_D = -\sum_{j>w}(-1)^j e_j(M)\varphi^{m-j}|_D$, a polynomial of degree at most $c(m-w-1) = c(\lceil K/c\rceil - 1) \le K-1$, so $c_M \in \mathrm{RS}[\mathbb{F},D,K]$, and U_{z^*} agrees with c_M on the cm fiber points of M. The map $M \mapsto c_M$ is injective on the pigeonhole class: equality of words of degree $\le K-1 \le n-1$ on all of D forces equality of polynomials, and the powers φ^{m-j} have distinct degrees $c(m-j)$, so all $e_j(M) = e_j(M')$ for $j > w$; for $j \le w$ they agree through z^*; hence $\prod_{b\in M}(Y-b) = \prod_{b\in M'}(Y-b)$ and $M = M'$. □

Theorem 3.2 (deep-point conversion)**.** *Let $C = \mathrm{RS}[\mathbb{F},D,k]$, $q > n$, let $\delta \in (0,1)$ satisfy $\lfloor \delta n\rfloor \le n-k-1$, and let $\eta \in (0,1)$. If $\varepsilon_{\mathrm{ca}}(C,\delta) \le \eta\,\frac{1}{k}\big(1-\frac{n}{q}\big)$, then every received word $U\colon D \to \mathbb{F}$ satisfies $\big|\Lambda\big(\mathrm{RS}[\mathbb{F},D,k+1],\delta,U\big)\big| \le q\,\varepsilon_{\mathrm{ca}}(C,\delta)/(1-\eta)$.*

Proof. Put $\varepsilon := \varepsilon_{\mathrm{ca}}(C,\delta)$ and $a := n - \lfloor\delta n\rfloor \ge k+1$, and let $P_1,\dots,P_L \in \mathbb{F}[X]_{<k+1}$ be distinct polynomials whose restrictions lie in $\Lambda(\mathrm{RS}[\mathbb{F},D,k+1],\delta,U)$; each agrees with U on a set S_i of at least a points, distances being integral. Let $\Omega := \mathbb{F}\setminus D$, of size $q-n$, and for $\alpha \in \Omega$ define the simple-pole pair $f_\alpha(x) := U(x)/(x-\alpha)$, $g_\alpha(x) := -1/(x-\alpha)$ for $x \in D$. For every i and every $x \in S_i$, $f_\alpha(x) + P_i(\alpha)g_\alpha(x) = (P_i(x) - P_i(\alpha))/(x-\alpha)$, the restriction of a polynomial of degree less than k; so the point at slope $P_i(\alpha)$ agrees with a codeword of C on S_i, hence is δ-close to C. The pair is δ-far from $C^{\equiv 2}$: if $G \in \mathbb{F}[X]_{<k}$ agreed with g_α on more than k points of D, then $(X-\alpha)G + 1$, of degree at most k, would have more than k roots while taking value 1 at α; so any pair explanation on $\ge a \ge k+1$ points is impossible. Therefore each distinct value among

$P_1(\alpha), \dots, P_L(\alpha)$ is a δ-CA-bad slope, and, writing $M(\alpha)$ for the number of distinct values, $M(\alpha) \le \varepsilon q$ for every $\alpha \in \Omega$.

It remains to pick a good α. For $i \neq j$ the nonzero polynomial $P_i - P_j$ has degree at most k, hence at most k roots, so the number of colliding pairs summed over $\alpha \in \Omega$ is at most $k\binom{L}{2}$; fix α whose collision count is at most $k\binom{L}{2}/(q-n)$. If the fibers of $i \mapsto P_i(\alpha)$ have sizes $m_1, \dots, m_{M(\alpha)}$, then $\sum_r m_r = L$ and $\sum_r m_r^2 = L + 2\sum_r \binom{m_r}{2} \le L + kL(L-1)/(q-n)$, so Cauchy–Schwarz gives $L^2 \le M(\alpha)\sum_r m_r^2$, i.e. $M(\alpha) \ge L(q-n)/(q-n+k(L-1)) \ge L(q-n)/(q-n+kL)$. Combining, $\varepsilon q(q-n+kL) \ge L(q-n)$, i.e. $L(q-n)(1-\varepsilon qk/(q-n)) \le \varepsilon q(q-n)$; the hypothesis gives $\varepsilon qk/(q-n) \le \eta < 1$, whence $L \le \varepsilon q/(1-\eta)$. □

Theorem 3.3 (cap criterion)**.** *In the setting of Lemma 3.1 with $K = k+1$ and $q > n$: if $\binom{N}{m} > |B|^w\left(\frac{q}{k}+1\right)$, then for every $\delta \in [1 - \frac{mc}{n}, 1-\rho)$,*

$$\varepsilon_{\mathrm{mca}}(C,\delta) \ \ge\ \varepsilon_{\mathrm{ca}}(C,\delta) \ >\ \frac{1}{2k}\Big(1 - \frac{n}{q}\Big).$$

Proof. Every $\delta < 1-\rho$ has $\lfloor \delta n \rfloor \le n-k-1$, since $\delta n < n-k$ and distances are integral; so Theorem 3.2 applies at every radius in the stated range. The word U of Lemma 3.1 has $\left|\Lambda(\mathrm{RS}[\mathbb{F}, D, k+1], \delta, U)\right| > \frac{q}{k} + 1 > \frac{q-n}{k}$ at every $\delta \ge 1 - \frac{mc}{n}$, because the $\binom{N}{m}/|B|^w$ listed codewords agree with U on $mc \ge (1-\delta)n$ points. If $\varepsilon_{\mathrm{ca}}(C,\delta) \le \frac{1}{2k}(1-\frac{n}{q})$, then Theorem 3.2 with $\eta = \frac{1}{2}$ bounds that list by $2q\,\varepsilon_{\mathrm{ca}}(C,\delta) \le \frac{q-n}{k}$, a contradiction. Lemma 2.2 transfers the bound to $\varepsilon_{\mathrm{mca}}$. □

Corollary 3.4 (ordinary locator cap; top of the plain-CA band)**.** *Let $D \subseteq \mathbb{F}$ be any set of size n, let $C = \mathrm{RS}[\mathbb{F}, D, k]$ with $\rho = k/n \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$, $k \le 2^{40}$ and $n \ge 2^{15}$, and assume $n < q < 2^{256}$. Define α_ρ and $s_\rho = \lceil \alpha_\rho n \rceil$ as in Theorem 1.1. Then, for every $\delta \in [1-\rho-s_\rho/n, 1-\rho)$,*

$$\varepsilon_{\mathrm{mca}}(C,\delta) \ge \varepsilon_{\mathrm{ca}}(C,\delta) > \frac{1}{2k}\Big(1 - \frac{n}{q}\Big).$$

In particular, if $q \ge 2^{128}$, then $\varepsilon_{\mathrm{ca}}(C,\delta) > 2^{-128}$ throughout this interval.

Proof. Apply Lemma 3.1 and Theorem 3.3 with the identity map $\varphi = X$, so $c = 1$, $N = n$, $B = \mathbb{F}$, $K = k+1$, $m = k + s_\rho$, and $w = s_\rho - 1$; here $k+1 \le m \le n$ because $s_\rho \ge 1$ and $\rho + \alpha_\rho + 2^{-15} < 1$ in all four cases. The cap criterion becomes

$$\binom{n}{k+s_\rho} > q^{s_\rho - 1}\Big(\frac{q}{k} + 1\Big).$$

It is enough to verify this with q replaced by 2^{256}. Since $k \ge n/16 \ge 2^{11}$, we have $\log_2(q/k+1) < 256$, while (1) gives

$$\log_2\binom{n}{k+s_\rho} \ge n\mathrm{H}_2\Big(\rho + \frac{s_\rho}{n}\Big) - \log_2(n+1).$$

Because $s_\rho/n \in [\alpha_\rho, \alpha_\rho + 2^{-15}]$, the following lower bounds hold on the relevant interval:

ρ	α_ρ	$\min \mathrm{H}_2(\rho + s_\rho/n)$	$\min \mathrm{H}_2(\rho + s_\rho/n) - 256\alpha_\rho$
$\frac{1}{2}$	$1/300$	0.9999	> 0.146
$\frac{1}{4}$	$3/1000$	0.8159	> 0.047
$\frac{1}{8}$	$41/20000$	0.5492	> 0.024
$\frac{1}{16}$	$3/2500$	0.3419	$> 0.034.$

Thus $n\mathrm{H}_2(\rho + s_\rho/n) - \log_2(n+1) > 256 s_\rho$ for every $n \ge 2^{15}$, with the tightest case still having more than 500 bits of slack at $n = 2^{15}$. This proves the criterion and hence the displayed lower bound. Finally, if $q \ge 2^{128}$ then $n \le 16k \le 2^{44}$, so $n/q < 2^{-84}$ and $\frac{1}{2k}(1-n/q) > 2^{-42} > 2^{-128}$. □

Corollary 3.5 (universal cap on the challenge box)**.** *Let $D \subseteq \mathbb{F}^\times$ be a coset of a cyclic subgroup of even order $n \ge 2^{15}$, let $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$ and $n < q < 2^{256}$, and set $g := g_\rho$ as in Theorem 1.1. Then $\varepsilon_{\mathrm{mca}}(C,\delta) > \frac{1}{2k}(1-\frac{n}{q})$ for every $\delta \in [1-\rho-g, 1-\rho)$.*

Proof. Since n is even and $n \mid q-1$, the field has odd characteristic and -1 is the element of order 2 in the subgroup H with $D = \alpha H$; so $\varphi := X^2$ has fibers $\{x, -x\}$ inside D, complete and of size 2. Apply Lemma 3.1 and Theorem 3.3 with $c = 2$, $B = \mathbb{F}$, $K = k+1$ and $m := \lceil (k+1+gn)/2 \rceil$; then $2m \geq k+gn$, so $[1-\rho-g, 1-\rho) \subseteq [1-\frac{2m}{n}, 1-\rho)$, and $w = m - \lceil (k+1)/2 \rceil \leq (gn+2)/2$, and $m \leq N = n/2$ since $k+gn+2 \leq n$. It remains to verify the criterion. Writing $\beta := \log_2 q \leq 256$, by (1) it suffices that

$$\frac{n}{2}\, \mathrm{H}_2(m/N) \;\geq\; 128\, gn + 512 + \log_2\!\left(\frac{n}{2} + 1\right), \tag{2}$$

since $\beta w \leq 128gn + 256$ and $\log_2(\frac{q}{k} + 1) \leq 256$. Here $m/N = 2m/n \in [\rho + g,\, \rho + g + 2^{-13}]$ for $n \geq 2^{15}$, and on that interval the concave H_2 is at least $h(\rho) := \min\big(\mathrm{H}_2(\rho+g), \mathrm{H}_2(\rho+g+2^{-13})\big)$, which evaluates (rounding down) to $h \geq 0.9999,\ 0.8122,\ 0.5489,\ 0.3410$ at $\rho = \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$. Both sides of (2) are linear in n up to the logarithm, with slopes $h/2$ and $128g$, and $h/2 - 128g$ equals $0.2499,\ 0.1561,\ 0.0244,\ 0.0455$ respectively, all positive; so it suffices to check (2) at $n = 2^{15}$, where the left side is at least $16384\, h \geq 8993$ (the tightest case, $\rho = \frac{1}{8}$) against a right side of $8192 + 512 + 15 = 8719$ there, and 5586 against $4096 + 527$ at $\rho = \frac{1}{16}$, and larger margins at $\rho \in \{\frac{1}{4}, \frac{1}{2}\}$; the slack of at least 274 bits makes the criterion of Theorem 3.3 strict. (All four instances were additionally verified by exact integer arithmetic at $n = 2^{15}$, $q = 2^{256} - 1$.) ☐

Proposition 3.6 (small fields)**.** *For every linear code $C \subsetneq \mathbb{F}^n$ and every $\delta \in (0,1)$, $\varepsilon_{\mathrm{mca}}(C, \delta) \geq 1/q$. Hence $\delta^*_C(\varepsilon^*) = 0$ whenever $\varepsilon^* < 1/q$, in particular whenever $q < 2^{128}$ at the grand target. At the boundary $q = 2^{128}$ the same construction gives $\varepsilon_{\mathrm{mca}} \geq 2^{-128}$ at every radius, which meets the threshold condition with equality and does not by itself force degeneracy.*

Proof. Pick $f_2 \notin C$, any $c_0 \in C$, $\gamma_0 \in \mathbb{F}$, and set $f_1 := c_0 - \gamma_0 f_2$. Then $f_1 + \gamma_0 f_2 = c_0$, so (c_0, D) is a proximity witness with $S = D$; but a pair explanation on $S = D$ would force $f_1, f_2 \in C$, which fails. Hence γ_0 is δ-MCA-bad for this pair at every δ, and $\varepsilon_{\mathrm{mca}}(C, \delta) \geq 1/q$. ☐

Remark 3.7 (other smooth domains). Lemma 3.1 and Theorem 3.3 are stated for arbitrary complete-fiber polynomial maps, so Corollary 3.5 extends beyond even-order multiplicative cosets with the identical arithmetic. (i) *Characteristic two, additive smoothness.* If $D = t + V$ is an $\mathbb{F}_2$-affine subspace of dimension $s \geq 1$ and $a \in V \setminus \{0\}$, the linearized polynomial $\varphi(x) = x^2 + ax$ satisfies $\varphi(x+a) = \varphi(x)$, so its fibers inside D are the pairs $\{x, x+a\}$, complete and of size 2; the proof of Corollary 3.5 then goes through verbatim. (ii) *Odd smooth orders.* If $3 \mid n$ (e.g. n a power of 3), take $\varphi = X^3$: the fibers are the μ_3-cosets, complete of size 3 since $\mu_3 \subseteq H$ and $3 \mid q-1$ forces characteristic $\neq 3$; the same entropy verification with $N = n/3$ yields the four gaps of Corollary 3.5 for $n \geq 2^{16}$ (checked as in the proof above, tightest margin again at $\rho = \frac{1}{8}$). (iii) Chebyshev and circle-code domains, and genus-one (ECFFT) domains, also carry complete uniform fibers and receive the same caps; see [Cho26b].

Remark 3.8 (small n and general rates). For $2^{12} \leq n < 2^{15}$ the corollary's uniform gap is not claimed, but capacity is still refuted: taking $m = \lceil (k+1)/2 \rceil + 1$, so $w = 1$ and agreement $2m \geq k+3$, the criterion $\binom{n/2}{m} > q(\frac{q}{k} + 1)$ holds at all four rates for $n \geq 2^{12}$ (at $\rho = \frac{1}{16}$, $n = 2^{12}$: $\log_2 \binom{2048}{130} \geq 687$ against at most 505), giving $\delta^*_C(2^{-128}) \leq 1 - \rho - 3/n$ whenever $q \geq 2^{128}$. For $n < 2^{12}$ with $q \geq 2^{128}$ our method gives nothing above the safe region, and we record this corner as genuinely open. For rates outside the four challenge values, Theorem 3.3 applies unchanged; the achievable gap is governed by the criterion, asymptotically the envelope of Proposition 7.1, and is positive for every fixed $\rho \in (0,1)$ and large n. Finally, the list challenge of [ABF26] is unsafe at the same first step: Lemma 3.1 at $K = k$ produces a single word with more than $2^{-128}q$ listed codewords under the analogous criterion, with the same verification.

4. The safe side

We now prove that MCA *holds*, with essentially optimal error, deep inside the unique-decoding region — for every linear code — and that up to half the minimum distance the mutual layer costs only an explicit tangent term.

Theorem 4.1 (deep regime)**.** *Let $C \subseteq \mathbb{F}^n$ be linear with minimum weight $w_{\min}$, let $\delta \in (0,1)$, $r := \lfloor \delta n \rfloor$, and assume $3r \le w_{\min} - 1$. Then for every pair (f_1, f_2) at least one of the following holds:*

(1) *there are $c_1, c_2 \in C$ and $T \subseteq [n]$ with $|T| \le r$ and $f_i = c_i$ off T for $i = 1, 2$; or*
(2) *at most $r+1$ slopes γ have $\operatorname{dist}(f_1 + \gamma f_2, C) \le \delta$.*

Consequently $\varepsilon_{\mathrm{ca}}(C, \delta) \le (r+1)/q$ and $\varepsilon_{\mathrm{mca}}(C, \delta) \le (r+1)/q$.

Proof. Suppose (2) fails, so there are $r+2$ distinct slopes $\gamma_0, \dots, \gamma_{r+1}$, codewords c_j, and error sets $E_j := \{i : (f_1 + \gamma_j f_2)(i) \ne c_j(i)\}$ with $|E_j| \le r$. For $j \ne l$, solving the two agreements off $E_j \cup E_l$ gives $f_2 = d_{jl} := (c_j - c_l)/(\gamma_j - \gamma_l)$ and $f_1 = e_{jl} := (\gamma_j c_l - \gamma_l c_j)/(\gamma_j - \gamma_l)$ off $E_j \cup E_l$, with $d_{jl}, e_{jl} \in C$. If $r = 0$ all E_j are empty and (1) holds with $T = \emptyset$; so assume $r \ge 1$, hence at least three slopes. For any three indices j, l, m, the codeword $d_{jl} - d_{jm}$ vanishes off $E_j \cup E_l \cup E_m$, a set of size at most $3r \le w_{\min} - 1$, so $d_{jl} = d_{jm}$; chaining through shared indices, all d_{jl} equal a single $d \in C$, and likewise all e_{jl} equal a single $e \in C$. Let $T := \{i : (f_1(i), f_2(i)) \ne (e(i), d(i))\}$. If $i \notin E_j$ and $i \notin E_l$ for two distinct indices, then solving at coordinate i gives $f_2(i) = d(i)$ and $f_1(i) = e(i)$, so $i \notin T$; contrapositively each $i \in T$ lies in at least $r+1$ of the $r+2$ sets E_j, whence $(r+1)|T| \le \sum_j |E_j| \le r(r+2)$ and $|T| \le r(r+2)/(r+1) < r+1$, i.e. $|T| \le r$. This is (1) with $(c_1, c_2) = (e, d)$.

For the error bounds, fix a pair. In case (2), both CA-bad and MCA-bad slopes are among the at most $r+1$ close slopes. In case (1): the pair is within column distance $r/n \le \delta$ of (c_1, c_2), so no slope is CA-bad. For MCA, write $\epsilon_i := f_i - c_i$, supported on T, and call γ *tangent* if $\epsilon_1(i) + \gamma \epsilon_2(i) = 0$ for some $i \in T$; there are at most $|T| \le r$ tangent slopes, since a coordinate with $\epsilon_2(i) \ne 0$ determines one value of γ and a coordinate with $\epsilon_2(i) = 0 \ne \epsilon_1(i)$ determines none. Let γ be non-tangent and let (c, S) be any proximity witness for $f_1 + \gamma f_2$, so $|S| \ge n - r$. The codeword $c - (c_1 + \gamma c_2)$ vanishes on $S \setminus T$, of size at least $n - 2r$, so its weight is at most $2r \le w_{\min} - 1$ and it is zero; then $(\epsilon_1 + \gamma \epsilon_2)|_S = 0$, and since γ is non-tangent this function is nonzero at every point of T, forcing $S \cap T = \emptyset$ and hence $f_i|_S = c_i|_S$: the witness extends mutually. So every MCA-bad slope is tangent, and there are at most $r \le r+1$ of them. ☐

Theorem 4.2 (mutual from correlated, below half the distance)**.** *Let $C \subseteq \mathbb{F}^n$ be linear with minimum weight $w_{\min}$, $\delta \in (0,1)$, $r := \lfloor \delta n \rfloor$, and assume $2r \le w_{\min} - 1$. Then $\varepsilon_{\mathrm{mca}}(C, \delta) \le \max\big(\varepsilon_{\mathrm{ca}}(C, \delta), r/q\big)$.*

Proof. Fix a pair (f_1, f_2). If $\operatorname{dist}_2((f_1, f_2), C^{\equiv 2}) > \delta$, every MCA-bad slope is CA-bad: proximity gives a witness, farness denies every pair explanation of size $\ge (1-\delta)n$, and the CA conditions are exactly these two; so the count is at most $q\, \varepsilon_{\mathrm{ca}}(C, \delta)$. If instead $\operatorname{dist}_2 \le \delta$, fix an explanation $(p_1, p_2) \in C^{\equiv 2}$ with column error set E, $|E| \le r$, and put $\epsilon_i := f_i - p_i$, supported on E. Exactly as in the proof of Theorem 4.1 — with $2r \le w_{\min} - 1$ now doing the forcing — for every non-tangent γ and every witness (c, S), the codeword $c - (p_1 + \gamma p_2)$ vanishes on $S \setminus E$ of size $\ge n - 2r$, hence is zero; then S avoids the support of $\epsilon_1 + \gamma \epsilon_2$, which is all of E, so $f_i|_S = p_i|_S$ and the witness extends. MCA-bad slopes are thus among the at most $|E| \le r$ tangent values. ☐

Corollary 4.3 (safe to half the distance, modulo one import)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$ and $\delta \le \frac{n-k}{2n}$. Assuming the unique-decoding proximity gap of* [BCIKS20, Thm. 4.1]*, namely $\varepsilon_{\mathrm{ca}}(C, \delta) \le n/q$ for δ up to half the minimum distance, we get $\varepsilon_{\mathrm{mca}}(C, \delta) \le n/q$.*

Proof. Here $w_{\min} = n - k + 1$ and $2\lfloor \delta n \rfloor \le n - k = w_{\min} - 1$, so Theorem 4.2 gives $\varepsilon_{\mathrm{mca}} \le \max(n/q, \lfloor \delta n \rfloor / q) = n/q$. ☐

5. Proof of Theorem 1.1

(1) is Proposition 3.6: $\varepsilon_{\mathrm{mca}}(C, \delta) \ge 1/q > 2^{-128}$ at every δ, so the supremum set is empty.

(2) By Corollary 3.4, $\varepsilon_{\mathrm{ca}}(C, \delta) > \frac{1}{2k}(1 - \frac{n}{q})$ for every $\delta \in [1 - \rho - s_\rho/n, 1 - \rho)$. Here $k \le 2^{40}$ gives $\frac{1}{2k} \ge 2^{-41}$, and $n \le 16k \le 2^{44}$ with $q \ge 2^{128}$ gives $1 - \frac{n}{q} > 1 - 2^{-84} > \frac{1}{2}$; so $\varepsilon_{\mathrm{ca}}(C, \delta) > 2^{-42} > 2^{-128}$ throughout the interval. By Lemma 2.2, the same interval is MCA-unsafe; hence no radius there belongs to the supremum set and $\delta^*_C(2^{-128}) \le 1 - \rho - s_\rho/n$. Finally $s_\rho/n \ge \alpha_\rho > g_\rho$, which places a nonempty CA-failing slice inside the old band $((1-\rho)/2, 1 - \rho - g_\rho)$.

(3) and (4). Let $r^* := \min(\lfloor (n-k)/3\rfloor, \lfloor 2^{-128}q\rfloor - 1)$ and suppose $r^* \geq 1$. At $\delta := r^*/n$ we have $\lfloor \delta n\rfloor = r^*$ and $3r^* \leq n-k = w_{\min}-1$, so Theorem 4.1 gives $\varepsilon_{\mathrm{mca}}(C,\delta) \leq (r^*+1)/q \leq 2^{-128}$, using $r^*+1 \leq \lfloor 2^{-128}q\rfloor$; also $\delta < 1-\rho$, so δ lies in the supremum set and $\delta^* \geq r^*/n$. If $q \geq 2^{128}n$ then $\lfloor 2^{-128}q\rfloor - 1 \geq n-1 \geq \lfloor (n-k)/3\rfloor$, so $r^* = \lfloor (n-k)/3\rfloor$ and $\delta^* \geq \frac{1}{n}\lfloor (n-k)/3\rfloor \geq \frac{1-\rho}{3} - \frac{2}{3n}$; since $s_\rho/n \geq g_\rho$, the earlier ratio estimate against the weaker upper bound $1-\rho-g_\rho$ still gives a factor smaller than three. With the import, take $\delta := \frac{1}{n}\lfloor (n-k)/2\rfloor \leq \frac{n-k}{2n}$: Corollary 4.3 gives $\varepsilon_{\mathrm{mca}} \leq n/q \leq 2^{-128}$ when $q \geq 2^{128}n$, so $\delta^* \geq \frac{1-\rho}{2} - \frac{1}{2n}$, and the same comparison gives a factor smaller than two. In the thin regime $2^{128} \leq q < 2^{128}n$ only the r^* clause is claimed; as $q \downarrow 2^{128}$, r^* reaches 0 and the lower bound becomes vacuous, as stated. □

Two comments on the statement. First, the import in (3) is a published theorem, so the two-sided determination within a factor of two stands on the literature; the unconditional clause records what the present paper proves alone. Second, the case split at $q = 2^{128}$ is genuinely necessary at the stated target: below it the threshold degenerates, at it neither degeneracy nor a nontrivial safe radius follows (Proposition 3.6), and the meaningful question there is posed at a larger target, as the deployed circle-code rows already illustrate [Cho26b].

6. Refining the residual band

The main theorem gives a two-sided pincer but does not by itself determine the exact staircase. The original broad residual questions can be sharpened further. First, the top of the old plain-CA band is already unsafe by the ordinary locator floor Corollary 3.4, which is stronger than the quadratic uniform cap used only to state the smooth-domain envelope. Second, the mutual layer above plain correlated agreement is not mysterious: for column-close pairs, every extra MCA slope is forced by either a tangent coordinate or a low-weight shortening image. A second, code-agnostic formulation bounds the same extra layer by a doubled-radius pair list.

6.1. **The top of the plain-CA band is already unsafe.** The gap g_ρ in Theorem 1.1 is the clean smooth-domain cap obtained from the two-fiber map X^2. The identity map gives a stronger top-edge statement for plain CA over arbitrary evaluation sets.

Corollary 6.1 (the original plain-CA band is not all safe)**.** *In the parameter box of Theorem 1.1, assume $q \geq 2^{128}$. Then*

$$\varepsilon_{\mathrm{ca}}(C,\delta) > 2^{-128} \qquad \textit{for every} \qquad \delta \in \left[1-\rho-\frac{s_\rho}{n},\, 1-\rho\right).$$

Since $s_\rho/n \geq \alpha_\rho > g_\rho$, the upper slice

$$\left[1-\rho-\frac{s_\rho}{n},\, 1-\rho-g_\rho\right)$$

of the previously displayed band is already plain-CA unsafe, and hence MCA unsafe.

Proof. This is exactly Corollary 3.4, followed by the numerical estimate in the proof of Theorem 1.1(2): $k \leq 2^{40}$ and $q \geq 2^{128}$ imply $\frac{1}{2k}(1-n/q) > 2^{-128}$. □

Thus the remaining plain-CA question starts below the sharper edge $1-\rho-s_\rho/n$, not below the coarser edge $1-\rho-g_\rho$.

6.2. **The mutual layer as a residual shortening image.** For a vector $u \in \mathbb{F}^D$ write $\mathrm{supp}(u) := \{x \in D : u(x) \neq 0\}$. Fix an integer r and a pair (f_1,f_2) that is r/n-close to $C^{\equiv 2}$. Choose one explaining pair $(p_1,p_2) \in C^{\equiv 2}$ and put

$$E := \{x \in D : (f_1(x),f_2(x)) \neq (p_1(x),p_2(x))\}, \qquad |E| \leq r,$$
$$e_i := f_i - p_i \qquad (i=1,2).$$

A slope is called *E-tangent* if

$$e_1(x) + \gamma e_2(x) = 0 \quad \text{for some } x \in E.$$

There are at most $|E| \leq r$ such slopes, because each coordinate with $e_2(x) \neq 0$ determines one finite value of γ, and a coordinate with $e_2(x) = 0 \neq e_1(x)$ determines none.

For this fixed explanation define $R_C(E, e_1, e_2; r)$ to be the set of slopes $\gamma \in \mathbb{F}$ for which there are a set $F_0 \subseteq D$ with $|F_0| \le r$ and a nonzero codeword $d \in C$ such that

$$\operatorname{supp}(d) \subseteq E \cup F_0, \qquad d(x) = e_1(x) + \gamma e_2(x) \quad \text{for every } x \in E \setminus F_0.$$

Finally put

$$R_C(r) := \max_{E, e_1, e_2} |R_C(E, e_1, e_2; r)|,$$

where the maximum is over $|E| \le r$ and error pairs supported on E.

Theorem 6.2 (shortening-image reduction for the mutual layer)**.** *Let $C \subseteq \mathbb{F}^D$ be linear, let $\delta \in (0, 1)$, and put $r := \lfloor \delta n \rfloor$. Then*

$$\varepsilon_{\mathrm{mca}}(C, \delta) \le \max\left(\varepsilon_{\mathrm{ca}}(C, \delta), \frac{r + R_C(r)}{q}\right).$$

If $C = \mathrm{RS}[\mathbb{F}, D, k]$, then every nonzero codeword counted in $R_C(r)$ belongs to a shortening of dimension at most

$$\max(0, 2r - (n - k)).$$

In particular $R_C(r) = 0$ whenever $2r \le n - k$, recovering the half-distance reduction Theorem 4.2.

Proof. Fix a pair (f_1, f_2). If the pair is δ-far from $C^{\equiv 2}$, then any MCA-bad slope is CA-bad by the proof of Lemma 2.2; this gives the first term.

Assume instead that the pair is δ-close, and choose (p_1, p_2) and E as above. Let γ be MCA-bad, with witness (c, S), $|S| \ge n - r$. Put $F_0 := D \setminus S$, so $|F_0| \le r$, and set

$$d := c - (p_1 + \gamma p_2) \in C.$$

On $S \setminus E$ the errors vanish and both c and $p_1 + \gamma p_2$ agree with $f_1 + \gamma f_2$, so $d = 0$ there. Hence $\operatorname{supp}(d) \subseteq E \cup F_0$. On $E \setminus F_0 = E \cap S$ we have

$$d = e_1 + \gamma e_2.$$

If γ is not E-tangent and $d = 0$, then $E \cap S = \emptyset$; consequently $S \subseteq D \setminus E$, and the original pair (p_1, p_2) explains (f_1, f_2) on S, contradicting MCA-badness. Thus every non-tangent MCA-bad slope belongs to $R_C(E, e_1, e_2; r)$, while tangent slopes contribute at most r. Taking the maximum over pairs proves the inequality.

For Reed–Solomon codes, fix E and F_0. The codewords supported in $E \cup F_0$ form a shortening on a set of size at most $2r$. Since Reed–Solomon codes are MDS, this shortening has dimension

$$\max(0, |E \cup F_0| - (n - k)) \le \max(0, 2r - (n - k)).$$

If $2r \le n - k$, every such shortening is zero, so no residual non-tangent slope exists. □

The next example shows that the shortening term is not an artifact of the proof: it appears as soon as the half-distance forcing condition fails.

Proposition 6.3 (sharpness of the half-distance wall)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, k]$ with minimum distance $d = n - k + 1$. Let*

$$\max(2, \lceil d/2 \rceil) \le r \le d - 2.$$

Then there is a pair (f_1, f_2) with $\operatorname{dist}_2((f_1, f_2), C^{\equiv 2}) \le r/n$ and a slope γ_0 that is MCA-bad at radius r/n but not CA-bad. Moreover γ_0 can be chosen non-tangent for a nearest explanation of the pair. Thus the estimate $\varepsilon_{\mathrm{mca}} \le \max(\varepsilon_{\mathrm{ca}}, r/q)$ cannot, in general, be extended beyond half the distance.

Proof. Choose disjoint sets $E, F_0 \subseteq D$ with $|E| = r$ and $|F_0| = d - r$, and put $W := E \cup F_0$, so $|W| = d$. Since C is MDS, there is a nonzero codeword $c \in C$ supported exactly on W; for Reed–Solomon codes one may take a nonzero scalar multiple of

$$\prod_{x \in D \setminus W} (X - x).$$

Let $S := D \setminus F_0$, so $|S| = n - d + r \ge n - r$ because $r \ge d/2$. Define f_1 to equal c on E and 0 off E. Choose f_2 supported on E whose restriction to E is not a scalar multiple of $c|_E$; this is possible because $|E| \ge 2$. The pair is r/n-close to $(0,0) \in C^{\equiv 2}$.

At slope $\gamma_0 = 0$, the word f_1 agrees with the codeword c on all of S: on E this is by definition, and on $S \setminus E$ both are zero. Hence (c, S) is a proximity witness. It is not a mutual witness. Indeed, any codeword agreeing with f_2 on S must be supported inside W; the shortening of an MDS code on a minimum-weight support W is one-dimensional and spanned by c, so its restriction to E is a scalar multiple of $c|_E$, contrary to the choice of f_2. Therefore γ_0 is MCA-bad. It is not CA-bad, because the pair is already r/n-close to $C^{\equiv 2}$. Finally, relative to the nearest explanation $(0,0)$, the value $f_1 + 0 \cdot f_2$ is nonzero at every point of E, so γ_0 is non-tangent. $\square$

6.3. **A doubled-radius pair-list formulation.** The shortening term can also be expressed without using the Reed–Solomon MDS structure. For a received pair $f = (f_1, f_2)$ and an integer s, define

$$\mathcal{L}_s^{(2)}(f) := \{(p_1, p_2) \in C^2 : \#\{x \in D : f_1(x) = p_1(x),\ f_2(x) = p_2(x)\} \ge n - s\},$$

and set

$$L_s^{(2)}(C) := \max_f |\mathcal{L}_s^{(2)}(f)|.$$

Theorem 6.4 (MCA excess is controlled by the doubled-radius pair list)**.** *Let $C \subseteq \mathbb{F}^D$ be linear, let $\delta \in (0,1)$, and put $r := \lfloor \delta n \rfloor$. Then*

$$\varepsilon_{\mathrm{mca}}(C, \delta) \le \max\left(\varepsilon_{\mathrm{ca}}(C, \delta), \frac{1 + (r+1)L_{2r}^{(2)}(C)}{q}\right).$$

Proof. Fix a pair $f = (f_1, f_2)$. If f is δ-far from C^2 in column distance, then every MCA-bad slope is already CA-bad, so this pair contributes at most $q\varepsilon_{\mathrm{ca}}(C, \delta)$ slopes.

Suppose therefore that f is δ-close to C^2. Let B be its set of MCA-bad slopes. If $B = \emptyset$ there is nothing to prove. Choose an anchor slope $\gamma_0 \in B$ and a non-extending witness (c_0, S_0), so $|S_0| \ge n - r$ and

$$f_1 + \gamma_0 f_2 = c_0 \quad \text{on } S_0.$$

For every $\gamma \in B \setminus \{\gamma_0\}$ choose a non-extending witness (c_γ, S_γ) and define

$$p_2^{(\gamma)} := \frac{c_\gamma - c_0}{\gamma - \gamma_0}, \qquad p_1^{(\gamma)} := \frac{\gamma c_0 - \gamma_0 c_\gamma}{\gamma - \gamma_0}.$$

By linearity, $P_\gamma := (p_1^{(\gamma)}, p_2^{(\gamma)}) \in C^2$. On $S_0 \cap S_\gamma$, whose size is at least $n - 2r$, the two line equations solve back to $f_1 = p_1^{(\gamma)}$ and $f_2 = p_2^{(\gamma)}$. Hence $P_\gamma \in \mathcal{L}_{2r}^{(2)}(f)$.

It remains to bound how many non-anchor slopes can map to the same pair $P = (p_1, p_2) \in \mathcal{L}_{2r}^{(2)}(f)$. Let

$$E_P := \{x \in D : (f_1(x), f_2(x)) \ne (p_1(x), p_2(x))\}, \qquad t := |E_P| \le 2r.$$

For every slope γ mapping to this same P, the corresponding witness codeword is $p_1 + \gamma p_2$, and on its witness set S_γ we need

$$(f_1 - p_1) + \gamma(f_2 - p_2) = 0.$$

Outside E_P this holds automatically. Inside E_P, a coordinate with $f_2(x) - p_2(x) \ne 0$ determines at most one value of γ, and a coordinate with $f_2(x) - p_2(x) = 0 \ne f_1(x) - p_1(x)$ determines none. Distinct finite slopes therefore have disjoint vanishing coordinate sets inside E_P.

If $t \le r$, then $D \setminus E_P$ has size at least $n - r$ and is a mutual extension set for P. A non-extending witness must include at least one coordinate of E_P at which the displayed tangent equation vanishes, so there are at most $t \le r$ such slopes. If $r < t \le 2r$, a witness of size at least $n - r$ must include at least $t - r$ coordinates of E_P at which the tangent equation vanishes. By disjointness, the number of such slopes is at most

$$\frac{t}{t - r} \le r + 1.$$

Thus each pair in the doubled-radius pair list receives at most $r+1$ non-anchor slopes. Adding back the single anchor gives $|B| \le 1 + (r+1)|\mathcal{L}^{(2)}_{2r}(f)|$, and the stated bound follows after maximizing over f and dividing by q. □

7. Barriers and what remains

The pincer above is not limited by lazy constants. Each mechanism stops for a structural reason: prefix floors have an entropy envelope, the deep-point conversion has an error ceiling, and the mutual layer has a genuine half-distance wall.

Proposition 7.1 (entropy envelope for prefix floors)**.** *In the setting of Lemma 3.1, write $\beta := \log_2|B|$, $A := mc$ and $g := (A-k)/n$, and let $K \le k+1$. If $\mathrm{H}_2(\rho+g) \le \beta\,(g - \frac{c}{n})$, then $\binom{N}{m} \le |B|^w$: the pigeonhole class is not guaranteed to exceed a single codeword, and no certificate of the form of Theorem 3.3 exists at agreement A and scale c. In particular, over any field with no proper subfield containing D and $q \ge 2^{128}$ (so $\beta \ge 128$), a certificate at scale c requires $g < \frac{1}{128} + \frac{c}{n}$; since a useful certificate also needs $\binom{N}{m} > \frac{q}{k} \ge 2^{88}$, hence $N \ge 89$ and $c \le n/89$, every certified gap is below $\frac{1}{128} + \frac{1}{89} < 2^{-5.6}$. At the ordinary scale $c = 1$ used in Corollary 3.4 this says the route cannot pass much beyond $1/128$; at the quadratic scale $c = 2$ used in Corollary 3.5 it gives the familiar $1/128 + 2/n$ envelope. For deployed rows whose domains lie in small subfields ($\beta = 31$) the same envelope, now $\mathrm{H}_2(\rho + g^*) = g^*\beta$, is wider, and it is reached and saturated in* [Cho26b].

Proof. Here $w = m - \lceil K/c\rceil \ge m - (k+c)/c = (A-k-c)/c$, so by (1), $\log_2\binom{N}{m} \le \frac{n}{c}\mathrm{H}_2(A/n) \le \frac{n}{c}\cdot\beta(g-\frac{c}{n}) = \beta\frac{gn-c}{c} \le \beta w$. The instantiation takes $\beta = \log_2 q \ge 128$ and $\mathrm{H}_2 \le 1$. □

Proposition 7.2 (ceiling of the conversion)**.** *Let δ be as in Theorem 3.2 and $T := \frac{1}{k}(1-\frac{n}{q})$. If some word u has $|\Lambda(\mathrm{RS}[\mathbb{F}, D, k+1], \delta, u)| \ge 1 + \kappa qT$ with $\kappa > 0$, then $\varepsilon_{\mathrm{ca}}(C,\delta) \ge \frac{\kappa}{\kappa+1}T$. Conversely, no list bound, however large, certifies $\varepsilon_{\mathrm{ca}} \ge T$ through Theorem 3.2: the certified error tends to T from below as $\kappa \to \infty$.*

Proof. Put $x := \varepsilon_{\mathrm{ca}}(C,\delta)$; if $x \ge T$ there is nothing to prove, so let $x < T$ and $\eta := x/T < 1$. Theorem 3.2 applies with this η and yields $\kappa qT \le qx/(1-x/T)$, so $\kappa(T-x) = \kappa T(1-x/T) \le x$, i.e. $x \ge \kappa T/(\kappa+1)$. The converse is read from the same inequality. □

Remark 7.3 (why half the distance is a wall)*.* The forcing step in Theorems 4.1 and 4.2 — a codeword vanishing outside a $2r$-set must be zero — fails precisely when $2r \ge w_{\min}$. For $C = \mathrm{RS}[\mathbb{F}, D, k]$ it fails with room to spare: codewords supported inside a fixed set W with $|W| = 2r$ are evaluations of polynomials of degree $< k$ vanishing on the $n - 2r$ points off W, a space of dimension exactly $\max(0, k-n+2r) = \max(0, 2r-(n-k))$. Proposition 6.3 shows that this is a real MCA obstruction, not merely a proof artifact.

The remaining tasks now have a narrower and more concrete form.

Problem 7.4 (middle plain correlated-agreement band)**.** Let $C = \mathrm{RS}[\mathbb{F}, D, k]$ be in the parameter box of Theorem 1.1 with $q \ge 2^{128}n$. Determine $\varepsilon_{\mathrm{ca}}(C,\delta)$ in the residual interval

$$1 - \sqrt{\rho} < \delta < 1 - \rho - \frac{s_\rho}{n}.$$

Equivalently, in the self-contained core without importing the current Johnson-radius CA theorems, replace the left endpoint by $(1-\rho)/2$. The original question below the coarse cap edge $1 - \rho - g_\rho$ is already negative on its upper slice by Corollary 6.1, positive for plain CA up to the Johnson radius by [BCIKS20, BCHKS25], and open only in the displayed middle range.

Problem 7.5 (residual shortening and pair-list images)**.** For $r = \lfloor \delta n \rfloor$ in the same middle interval, bound the residual term $R_C(r)$ of Theorem 6.2, or equivalently the doubled-radius pair-list quantity $L^{(2)}_{2r}(C)$ of Theorem 6.4. A target-level bound

$$r + R_C(r) \le 2^{-128}q$$

together with a solution of Problem 7.4 would close the MCA threshold in the middle band. A polynomial bound $L^{(2)}_{2r}(C) \le n^A$ gives the same conclusion for fields satisfying $q \ge 2^{128} n^{A+1}$ whenever plain CA is safe. In the companion framework [Cho26b], periodic witnesses are counted by quotient and extension ledgers; what remains is the aperiodic Hankel/residue-line packing problem.

Thus the current complete picture is this. Small fields are degenerate. Large fields are safe to one third of the distance by a self-contained theorem and safe to one half of the distance with the single BCIKS import. They are unsafe, already for plain correlated agreement, from $1 - \rho - s_\rho/n$ to capacity; over deployed subfield rows, optimized companion floors push this unsafe frontier much farther left. What remains is not the original broad pair of problems, but the middle CA interval of Problem 7.4 and the residual shortening or pair-list kernel of Problem 7.5.

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